Venkata Mani Sai Praneeth Chidella

+1 716-306-8906 | chsaipraneeth0@gmail.com | LinkedIn | **Open to relocation**

Professional Summary

AI Engineer with 4+ years of experience building production-grade LLM systems, multi-agent pipelines, and scalable ML infrastructure. Delivered measurable outcomes including 20%+ reduction in operational costs, sub-100ms inference latency, and automated reasoning workflows across fraud detection, claims processing, and NLP classification tasks. Hands-on across the full ML lifecycle from data engineering and model training to MLOps, evaluation (RAGAS/G-Eval), and deployment using GPT-4o, Claude 3.5 Sonnet, and local SLMs. Requires H-1B sponsorship.

Skills

Languages & Tools: Python (Advanced), Java, JavaScript, SQL, R, MATLAB, Jupyter Notebook, VS Code, Google Colab
AI & LLM Orchestration: LangGraph, LangChain, Multi-Agent Systems, GPT-4o/Vision, Claude 3.5 Sonnet, Ollama, Prompt Engineering (CoT, ReAct), Structured Output Validation (Pydantic), LangSmith
Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, SVM, Feature Engineering, SMOTE/ADASYN, Model Drift Monitoring, Explainable AI (SHAP, LIME)
Deep Learning & NLP: PyTorch, CNN, LSTM, Transformers, BERT, LLaMA, Hugging Face, LLM Fine-Tuning, Computer Vision
Retrieval & Evaluation: RAG Pipelines, Vector Databases (Pinecone, Milvus, ChromaDB), Semantic Chunking, Hallucination Detection, RAGAS, G-Eval, Token/Cost Optimization
Cloud & MLOps: AWS (S3, EC2, Lambda, SageMaker, Glue, Redshift, DynamoDB), GCP Vertex AI, Azure DevOps, Docker, Kubernetes, MLflow, FastAPI, CI/CD, Async Programming
Data & Visualization: Pandas, NumPy, PySpark, Spark Streaming, Hadoop, PostgreSQL, MongoDB, Redis, Tableau, Power BI, Streamlit, Matplotlib, Seaborn
Statistical Techniques: Hypothesis Testing, A/B Testing, Causal Inference, Time Series Forecasting (ARIMA, LSTM), Anomaly Detection, Experimental Design
Operating Systems: Windows, Linux, macOS

Work Experience

Walmart, USA — **Aug 2024 – Present**
AI Engineer

- Built LLM-based search and personalization pipelines for product discovery, reducing average search latency by 35% and improving click-through rates by 18% across the e-commerce platform.
- Developed ML models for inventory management using historical sales and real-time demand signals, reducing stockout incidents by 22% and cutting excess inventory holding costs by 15%.
- Set up MLOps workflows on Vertex AI and Kubernetes for automated model retraining and deployment, reducing manual redeployment effort by 70% and maintaining 99.8% uptime for production AI services.
- Built collaborative filtering and embedding-based recommendation systems, contributing to a 12% lift in conversion rate and a 9% increase in average session engagement.
- Designed reinforcement learning agents for route and logistics optimization, cutting average delivery time by 11% and reducing fuel-related costs by 8% across the distribution network.
- Built a computer vision solution using YOLO and OpenCV for real-time retail shelf monitoring — detecting out-of-stock products, misplaced items, and compliance issues — deployed on edge AI devices with NVIDIA CUDA, reducing manual shelf audit time by 45%.
- Collaborated with product and data engineering teams to define evaluation metrics and monitor model performance post-deployment, ensuring outputs stayed aligned with business goals.

Maxgen Technologies, India — **Dec 2021 – Dec 2023**
AI Engineer

- Engineered a real-time fraud detection engine using an ensemble of Random Forest and XGBoost, processing 10,000+ daily transactions at 94% precision.
- Built a high-dimensional feature pipeline using SQL and Pandas to extract 50+ behavioral indicators, improving detection of sophisticated fraud patterns by 20%.
- Resolved extreme class imbalance using cost-sensitive learning and SMOTE/ADASYN, achieving 0.89 AUC-ROC on production transaction data.
- Integrated Explainable AI modules using SHAP and LIME, generating human-readable reasoning codes for flagged transactions to streamline manual audit workflows.
- Deployed fraud models as Dockerized FastAPI microservices, maintaining sub-100ms response times under peak load.
- Built an automated MLOps loop with MLflow for versioning and CI/CD for retraining, keeping model performance stable as fraud tactics evolved.
- Implemented drift monitoring and A/B testing of decision thresholds, reducing long-term performance decay and false negatives by 60% over 12 months.

- Built an intelligent document processing pipeline using OCR (Tesseract), Hugging Face Transformers, and Azure Cognitive Services to extract and classify information from invoices and contracts, achieving 91% extraction accuracy and cutting manual data entry by 55%.

Smart Claim AI, USA **Jan 2024 – Jul 2024**
AI Engineer (Volunteer / Startup)

- Built the full-stack foundation of an AI-powered insurance claims platform using React, Node.js/Express, and MongoDB Atlas, deployed on Vercel and Render with OAuth authentication (Google, Facebook, GitHub).
- Integrated GPT-4 Vision to analyze uploaded damage images — identifying damage type, affected areas, and severity — and connected it with the Claude API to generate structured, adjuster-ready claim reports.
- Built a RAG pipeline to handle jurisdiction-specific insurance rules that vary by state and county, grounding all AI outputs in verified policy documents and preventing hallucinations in sensitive claim decisions.
- Designed the AI pipeline to produce complete claim summaries including damage descriptions, severity estimates, recommended next steps, and draft documentation — reducing the manual workload for adjusters.
- Orchestrated two AI APIs (OpenAI GPT-4 Vision + Anthropic Claude) within a single production pipeline, combining multimodal image analysis with language-based report generation.
- Secured all AI outputs by restricting responses to verified regulatory and policy documents via RAG, ensuring no incorrect or sensitive information was surfaced to end users.

Education

Master of Science in Computer Science | State University of New York at Buffalo, Buffalo, NY May 2025
Relevant Courses: Machine Learning, Deep Learning, Computer Vision and Image Processing, Data Intensive Computing, Statistical Learning and Data Mining, Data Models and Query Language, Programming and Database Fundamentals for Data Science, Operating Systems and Computer Security.

Projects

- **Credit Card Fraud Detection** | ***Python, Scikit-learn, Random Forest, Isolation Forest, Logistic Regression, Tableau***
  Built an anomaly detection pipeline combining supervised and unsupervised methods to catch fraudulent transactions. Improved fraud recall by 22% while keeping false positives low. Connected the model output to a real-time Tableau dashboard for live monitoring and alerts, giving operations teams visibility into flagged transactions as they happen.
- **Diabetes Risk Prediction** | ***PySpark, Python, Scikit-learn, Streamlit***
  Trained a distributed Random Forest and Gradient Boosted Tree pipeline on patient health data, achieving 99.72% accuracy while cutting training time by 40% through PySpark parallelism. Deployed the model as an interactive Streamlit app where users can input health metrics and receive a risk score instantly, with automated PDF report generation for clinical documentation.
- **CrimeScope** | ***Python, Flask, LangChain, Ollama / Mistral 7B, FAISS, Scikit-learn***
  Built a fully local RAG-based crime analysis chatbot on Buffalo crime data using FAISS vector search and Mistral 7B via Ollama. Zero cloud dependency; sub-second query latency with strict instruction-based response constraints to reduce hallucinations.
- **Local RAG & Privacy-First QA System** | ***Python, Ollama, ChromaDB, Semantic Chunking***
  Architected a fully local RAG pipeline using Ollama and ChromaDB to eliminate cloud data privacy risks and API latency. Implemented semantic chunking and cosine-similarity retrieval for contextually relevant responses.
- **OAuth Authentication System** | ***React, Node.js, Vercel, Render***
  Built and deployed a full-stack OAuth authentication system supporting Google, Facebook, and GitHub login, with React frontend on Vercel and Node.js backend on Render.

Research & Publication

IEEE – Telugu Idiomatic Sentence Classification **Published**
Published – IEEE (2023) | View Publication
Developed a deep learning classification model for Telugu idiomatic expressions achieving 98% accuracy. Addressed the challenge of low-resource NLP for regional Indian languages using custom feature engineering and neural classification architectures.

Detection of Offensive Text in Memes Using Deep Learning Techniques **Published**
ResearchGate (2024) | View Publication
Developed a deep learning pipeline to detect offensive and hateful text embedded in meme images, combining computer vision and NLP techniques for multimodal classification.